Jnana

Layered Constitutional Containment for Governed Computation

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Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

Jnana is a candidate governed semantic compilation architecture that separates canonical meaning, admissibility law, constitutional amendment constraints, procedural execution, external effects, structural receipts, and derived projections. Its central question is not merely whether computation can run, but whether a proposed transition may be admitted, lowered, executed, allowed to affect a domain, committed to canonical state, and retained as precedent under declared authority and evidence.

The source paper presents this architecture through a biological metaphor: MCIR as semantic genome, MCIR-Law as developmental law, PEIR as regulated expression, Domain Effects as an immune membrane, ReceiptLog as structural memory, and state projections as phenotype. The metaphor is pedagogically strong, but it is not evidence of implementation or conformance. This revision makes the boundary contracts explicit, separates architecture from implementation, qualifies claims of semantic identity and negative-power guarantees, adds ambiguous-effect and recovery states, and locates Jnana within the larger Governed Semantic Kernel, Constitutional Computing, Governed Meaning Base, and semantic-language stack.

Jnana's strongest contribution is **layered constitutional containment**. A bounded semantic root and decidable admission regime can govern potentially expressive execution only if execution cannot silently widen meaning or authority, external effects cross explicit membranes, receipts remain independently checkable, and every conformance claim is limited to the verified vertical slice. Refusal is a valid outcome. So are deferment, explicit gap, and unresolved effect status. Persistence is earned by evidence; it is never inferred from narrative continuity.

1. Definition, Scope, and Claim Ceiling

Jnana is a candidate architectural configuration of a Governed Semantic Kernel in which semantic proposals move through distinct representation, law, execution, effect, evidence, and projection boundaries [JNANA-II].

The architecture proposes these roles:

1. **MCIR**: canonical governed meaning and transformation structure;
2. **MCIR-Law**: a bounded, versioned admission regime;
3. **Constitutional Invariants**: constraints on law change and authority widening;
4. **PEIR**: procedural lowering and expressive execution plans;
5. **Domain Effects**: capability- and jurisdiction-bounded crossings into external domains;
6. **ReceiptLog**: independently checkable structural memory;
7. **State Projection**: derived, reproducible views for people and agents.

The source and broader corpus also contain specifications, codebases, staged runtime packages, courts, replay reports, and integration candidates carrying the Jnana name. Their presence is evidence of an implementation program, not automatic proof that one current artifact conforms to this paper or realizes the whole architecture. Exact implementation identity, version-family boundaries, and conformance remain separately adjudicated.

Jnana is not presently established here as:

- a universally correct constitution;
- a complete implementation or released operating system;
- a proof that semantic equivalence is decidable in every domain;
- a guarantee of justice, freedom, safety, correctness, or non-coercion;
- a replacement for law, institutional authority, user consent, or domain evidence;
- permission for any implementation, deployment, effect, or binding;
- a claim that biological systems literally instantiate the architecture.

2. The Biological Metaphor and Its Boundary

The source calls Jnana **computer DNA with admissibility**. The analogy organizes the architecture:

- **Jnana concern:** MCIR
  Biological metaphor: genome
  Architectural obligation: canonical, typed, versioned meaning-bearing structure
- **Jnana concern:** MCIR-Law
  Biological metaphor: developmental law
  Architectural obligation: explicit admission rules and outcomes

- **Jnana concern:** Constitutional Invariants
  Biological metaphor: inherited viability constraints
  Architectural obligation: bounded amendment and authority widening
- **Jnana concern:** PEIR
  Biological metaphor: regulated expression
  Architectural obligation: lowering that cannot invent commitments silently
- **Jnana concern:** Domain Effects
  Biological metaphor: membrane and immune response
  Architectural obligation: explicit capability and effect jurisdiction
- **Jnana concern:** ReceiptLog
  Biological metaphor: structural memory
  Architectural obligation: tamper-evident evidence and lineage
- **Jnana concern:** State Projection
  Biological metaphor: phenotype
  Architectural obligation: derived views tethered to exact state and history

The metaphor helps readers see why representation, expression, effect, and memory should not collapse into one runtime. It should not carry proof. Biological evolution does not establish that Jnana's rules are correct, that a refusal is healthy, or that a mutation is illegitimate. Terms such as *cancer*, *apoptosis*, *immune*, and *organism* remain explanatory analogies unless a separately defined computational property replaces them.

The operative architectural question is therefore:

> Is this proposed transition admissible under the exact law, authority, evidence, state, and effect boundary selected for this Jnana instance?

That question is narrower and more testable than asking whether a meaning is "allowed to exist." Material that cannot enter canonical Jnana state may still exist as source, residual, rejected proposal, external fact, or unresolved gap. Non-admission must not erase it.

3. Architecture Versus Instance

The word **Jnana** currently spans several identity levels that must remain distinct:

- architectural concept;
- selected five-layer Governed Semantic Kernel configuration;
- specifications and constitutional codices;
- compiler, runtime, broker, court, or service candidates;
- deployment instances and their state;
- evidence packages, receipts, and review reports;
- pedagogical metaphors and paper claims.

A document with Jnana in its title is not thereby the architecture. A runtime sharing a source tree is not thereby a conforming implementation. A passing local court does not establish corpus-wide or cross-host conformance. A deployed instance is not identical to its specification or version family.

Every conformance statement should therefore identify:

$$\begin{aligned} Claim = \langle &architectureVersion, profile, instanceConfiguration, \\ &artifactSet, environment, court, evidence, claimCeiling \rangle. \end{aligned}$$

Without these coordinates, "Jnana works" is not a testable claim.

4. The Layered Transition Model

Let a configured Jnana instance be:

$$J = \langle M, L, K, P, E, R, Q \rangle,$$

where:

- M is the MCIR grammar, canonicalization, and semantic root;
- L is the selected MCIR-Law admission regime;
- K is the constitutional and amendment kernel;
- P is the PEIR lowering and execution regime;
- E is the Domain Effect membrane and capability set;
- R is the ReceiptLog and external evidence interface;
- Q is the state-projection regime.

A proposed transition follows partial, independently receipted stages:

$$Canon_M(source) \rightharpoonup m \mid GAP,$$

$$\begin{aligned} &Admit_{L,K}(m, state, authority, evidence) \rightharpoonup ADMIT \\ &\mid DEFER \\ &\mid REFUSE \\ &\mid GAP \end{aligned},$$

$$Lower_P(m, decision) \rightharpoonup p \mid GAP,$$

$$Run_P(p) \rightharpoonup intents \mid EXECUTION_UNKNOWN,$$

$$
\begin{aligned}
Cross_E(intent) \rightharpoonup\ & EFFECT_COMMITTED \\
\mid\ & EFFECT_REFUSED \\
\mid\ & EFFECT_UNKNOWN \\
\mid\ & RECONCILIATION_REQUIRED
\end{aligned}
.
$$

Canonical state may record only what its evidence supports. A missing receipt cannot undo an effect that may already have occurred. It requires an unknown or reconciliation state, not a convenient claim that nothing happened.

The model makes the architecture's central separation explicit:

```
source or proposal
  -> canonical candidate
  -> admission decision
  -> procedural lowering
  -> execution and effect intents
  -> domain-effect decisions and outcomes
  -> canonical commit or explicit ambiguity
  -> independent receipt lineage
  -> derived projections
```

5. MCIR: Canonical Meaning Without Semantic Omniscience

MCIR provides a closed, versioned structural locus for governance-relevant meaning [MCIR]. It can make a change to admissibility, authority, transformation semantics, or capability exposure mechanically visible.

Three source claims require precision.

First, hash equality establishes identity of canonical bytes under an exact grammar, schema, normalization regime, and version. It does not by itself establish semantic equivalence across grammar versions, profiles, contexts, or interpretation regimes.

Second, canonicalization does not eliminate all ambiguity. It may represent ambiguity, alternatives, unresolved references, gaps, and nonclaims explicitly. If source meaning cannot be represented without loss, the compiler must retain the source and emit a residual or gap record rather than declare non-existence.

Third, MCIR does not decide which governance is correct. It makes governance-bearing structure inspectable. Admission remains a separate operation.

Minimum MCIR obligations within Jnana include:

- exact grammar and schema identity;
- deterministic encoding and normalization;
- source and residual custody;

- semantic identity and reference resolution;
- governance-relevance rules;
- canonical test vectors and independent implementations;
- explicit migration across versions.

6. MCIR-Law and Constitutional Amendment

MCIR-Law evaluates proposals under an exact law head, jurisdiction, state, authority context, evidence bundle, and time. It should emit `ADMIT`, `DEFER`, `REFUSE`, or `GAP` with stable reason codes and a decision receipt.

A named law does not validate itself. The root of trust, authority registry, delegation chain, revocation state, and applicable jurisdiction remain external dependencies that must be bound and checked. A proposal cannot manufacture the authority that admits it.

Law change also needs a non-circular transition:

1. encode a candidate successor law under the current semantic grammar;
2. evaluate the amendment under the predecessor law and constitutional kernel;
3. bind the authorized actors, evidence, migration, and effective epoch;
4. activate the successor as a new law head;
5. retain the predecessor, decision receipt, incompatibilities, and rollback or recovery limits.

This is not "law judging itself" in the abstract. It is one exact predecessor regime deciding whether it authorizes a named successor transition. The initial constitutional root remains a declared boot commitment, not something the system can prove legitimate from within itself.

Constitutional Invariants are therefore entrenched constraints, not magical immutable truths. If they can change, the architecture must name the exceptional successor-epoch process. If they cannot change, the implementation must state what preserves that immutability and how incompatible communities or users may exit, fork, or reject the regime. Declared immutability does not itself establish legitimacy.

7. PEIR and Expressive Containment

PEIR is the procedural execution layer. It may be substantially more expressive than MCIR-Law, including Turing-complete computation, while remaining contained by decidable admission and effect boundaries [TAPD].

PEIR must not silently:

- invent meaning absent from the admitted artifact;
- widen authority, capability, jurisdiction, or effect scope;
- replace an admitted dependency or law head;

- reinterpret a refusal or gap as permission;
- suppress a residual, exception, or nonclaim;
- treat execution success as evidence of semantic correctness.

Lowering therefore needs a correspondence obligation. The lowered plan binds to the admitted semantic artifact, law decision, compiler identity, compiler configuration, target environment, and expected effect surface. Independent or diverse-build checks may strengthen evidence, but no compiler can prove its own trustworthiness merely by producing a receipt.

The precise relation among PEIR, GCML, GOML, GSOL, and other procedural or operational representations remains an author decision. A safe provisional distinction is that GOML represents governed objects and roles, GSOL represents governed operations, GCML proposes a machine-facing procedural contract, and PEIR names an execution-oriented lowering within Jnana. This is a comparison map, not an identity merger or final universal pipeline [GSOL] [GCML].

8. Domain Effects and Effect Ambiguity

Domain Effects form the boundary between computed intent and external consequence. Every effect request should identify:

- effect type and target;
- required capability and authority;
- jurisdiction and applicable law;
- preconditions and expected prior state;
- idempotency or replay class;
- timeout and ambiguity policy;
- evidence provider and trust class;
- compensation, rollback, or reconciliation path;
- result and receipt bindings.

An attested receipt supports a claim about an effect; it is not identical to the effect. An effect can occur without a durable receipt, a receipt can be wrong or forged, and an external system can return success without satisfying the intended semantic condition. Jnana must therefore represent `EFFECT_UNKNOWN` and

`RECONCILIATION_REQUIRED` rather than forcing every call into success or refusal.

Authority at the membrane should be explicit, minimal, scope-bound, time-bound, revocable, and non-transitive. The execution environment cannot inherit ambient authority merely because the program was admitted.

9. ReceiptLog as Externalized Structural Memory

ReceiptLog preserves proposed transitions, decisions, lowerings, effect attempts, observations, commits, refusals, gaps, contests, and revisions. Its most important property is not narrative completeness but independent checkability.

Minimum obligations include:

- append-only or equivalently tamper-evident lineage;
- exact bindings to subject, law, meaning, state, authority, evidence, environment, and time;
- privilege separation from the process being audited;
- stable event identity and ordering semantics;
- explicit replay class and nondeterminism disclosures;
- crash, truncation, duplication, and equivocation handling;
- external export and verification;
- retention, privacy, redaction, and selective-disclosure rules.

A receipt proves only what its evidence path supports. It cannot transform a bad law, stale authority, incomplete observation, or misconfigured verifier into a correct outcome [RWR] [THR].

10. Projection Is Not Truth

State projections are derived views over canonical state, receipts, evidence, and selected interpretation. They should bind to exact roots, projection code, configuration, time, and freshness.

A projection is discardable only if it is reproducible from retained authoritative inputs. If the inputs are unavailable, private, mutable, or nondeterministic, discardability is not assured. Interfaces must expose stale, partial, conflicted, and non-reproducible status rather than presenting a smooth view as truth.

The **Governed Meaning Base (GMB)** is the current name in the historical `CCF -> CMB -> GMB` lineage. GMB can retain Jnana specifications, semantic artifacts, laws, decisions, receipts, conflicts, evidence, implementation claims, and projections. It is a governed meaning and knowledge substrate, not automatically a Jnana runtime or admission authority. Hyperpedia can project these objects pedagogically while preserving custody and standing.

11. Negative Power and the Ableman Prime Directive

The source connects Jnana to the Ableman Prime Directive (APD): affected agents should be able to detect bindings, refuse or exit where refusal is lawful, and renegotiate applicability unless immutability is explicit [TAPD]. These capacities preserve **negative power**: the practical ability to prevent, withhold, refuse, contest, exit, or revise a binding.

Let a binding record be:

$$B = \langle subject, object, rule, scope, authority, conditions, \\ visibility, refusal, exit, contest, renegotiation, time \rangle.$$

An APD-oriented conformance claim must show more than fields in a schema. It must test whether the affected agent can actually:

- discover which bindings apply;
- understand their source and consequence;
- invoke refusal, contest, or exit through reachable interfaces;
- avoid silent retaliation or catastrophic technical lock-in within the system boundary;
- carry material objections into successor decisions;
- determine when a binding is claimed to be immutable and challenge whether that claim applies.

Jnana alone cannot guarantee absence of external legal, economic, physical, or social coercion. It can expose and constrain bindings inside a declared socio-technical boundary, produce evidence of available pathways, and refuse to claim APD conformance where those pathways are absent or merely performative.

Explicit immutability is not enough. The authority to declare a binding immutable, the affected scope, the exit or fork options, and the successor-epoch path must also be visible. Otherwise "immutable" becomes a label for unreviewable power.

12. Refusal, Deferment, Gap, and Recovery

Refusal is an integrity-preserving outcome when the named law establishes that a proposal is inadmissible. It should include scope, reasons, evidence, law version, authority, time, and available escalation or revision paths.

Not every non-success is refusal:

- `DEFER` means specified evidence or authority may later resolve the decision;
- `GAP` means the representation, identity, law, or target is insufficient to decide;
- `EXECUTION_UNKNOWN` means the procedural result cannot be established;
- `EFFECT_UNKNOWN` means an external consequence may have occurred without adequate evidence;
- `RECONCILIATION_REQUIRED`
  means canonical state must not pretend certainty until bounded recovery completes.

The biological term **apoptosis** is useful only as an analogy for an orderly, integrity-preserving stop. A refused proposal is not diseased, immoral, or false. It is inadmissible under one bounded regime and may remain available for correction, appeal, fork, or evaluation elsewhere.

13. Emergency and Break-Glass Semantics

"No temporary override" and "break glass" conflict unless emergency authority is defined carefully.

A constitutional break-glass path should be a predeclared capability with:

- narrow trigger conditions;
- named holders and delegation limits;
- exact effect scope and duration;
- unavoidable logging where technically possible;
- explicit ambiguity handling where logging may fail;
- automatic expiry and privilege contraction;
- recovery-epoch entry;
- independent post-event review;
- successor-state and restitution obligations.

If emergency action bypasses these rules, it is not a lawful Jnana transition merely because a later receipt describes it. The system should preserve the event as an exception or violation and must not retroactively launder it into ordinary authority.

14. Incremental Release and Honest Conformance

The source states that Jnana cannot be partially released. That is too strong. Components, specifications, prototypes, and bounded vertical slices can be released incrementally when their standing and omissions are explicit.

What cannot be partial is the **claim of complete Jnana conformance**. A release that lacks an effect membrane, independent receipts, amendment semantics, recovery states, or usable refusal paths must not imply those properties. Each release should declare:

- which layers and contracts are present;
- which profiles and configurations are supported;
- which threats and failure states were tested;
- which evidence is reproducible;
- which layers are simulated, stubbed, external, or absent;
- the maximum claim the evidence permits.

A minimal conformance candidate should be a complete low-consequence vertical slice rather than a large collection of unbound components.

15. Relation to Constitutional Computing

Jnana is one technical realization pathway within the broader proposal for Constitutional Computing, not the whole field [CC]. Its five-layer containment pattern can support sovereign and inspectable software by making semantic commitments, authority, lowering, effects, and evidence separable [SS]. Constraint-Bearing Composability supplies a test for whether obligations survive composition across those layers [CBC]. Jnana II extends the architecture toward a constitutional transformation substrate and should be reconciled as a successor or companion work rather than silently merged [JNANA-II].

The current corpus suggests this provisional division:

```
Constitutional Computing   broad computing paradigm
Governed Semantic Kernel substrate-neutral constitutional kernel
pattern
Jnana selected layered architecture/configuration family
MCIR / MCIR-Law            bounded semantic root and admission regime
PEIR / GCML / GSOL / GOML adjacent execution, machine, operation,
and object surfaces
Domain Effects             effect-jurisdiction membrane
ReceiptLog                 evidence and lineage surface
GMB                        governed meaning/knowledge substrate
MosaicOS user-facing sovereign computing realization family
```

Every relation remains versioned and evidence-bound. Shared vocabulary or source-tree proximity does not establish identity or conformance.

16. Configuration Manifold

Jnana admits many possible configurations:

- centralized, federated, personal, institutional, and peer-to-peer authority;
- immutable-root, amendable-root, epochal, forkable, and plural constitutional regimes;
- deterministic, bounded-nondeterministic, replayable, and externally reconciled execution;
- capability, object, operation, role, and jurisdiction-based effect control;
- local, remote, hardware-bound, virtualized, and cross-substrate realization;
- public, private, selectively disclosed, and privacy-preserving receipts;
- strict, permissive, multi-court, appealable, and community-specific admission.

CM1 contains every configuration satisfying the declared architecture and minimum layer contracts. It may still contain centralized, brittle, exclusionary, slow, expensive, surveillance-heavy, or capture-prone designs.

CM2 prefers configurations that reduce ambient authority, single-writer capture, irrecoverable ambiguity, semantic loss, forced disclosure, catastrophic exit cost, and unverifiable claims while improving substitution, recovery, accessibility, privacy, forkability, independent verification, and user sovereignty.

Instance Configuration selects exact grammar, law, constitutional root, authority registry, compiler, runtime, effect adapters, capabilities, receipt store, projection code, environment, trust anchors, recovery policy, and conformance court.

Jnana as Layered Constitutional Containment Architecture

The completed P1 corpus supports a compressed role assignment for Jnana. Jnana is the **layered constitutional containment architecture** that relates meaning representation, law, procedural lowering, effect mediation, receipts, and projection. It should not absorb the identities of its formats, kernels, enforcement profiles, implementation manuals, or successor proposals.

- **Adjacent work:** *Ableman Format Specification*
  Supplies: Specification ecosystem and separations among domain physics, configuration, instance selection, realization, evidence, and revision.
  Jnana boundary: AFS specifies artifacts and profiles; it is not automatically the Jnana runtime.
- **Adjacent work:** MCIR family
  Supplies: Canonical Intermediate Representation for admitted meaning and obligations.
  Jnana boundary: MCIR is a representation layer, not the whole architecture or its authority source.
- **Adjacent work:** *The Implementation Codex*
  Supplies: Construction-facing translation from specification into bounded work and evidence.
  Jnana boundary: Implementation guidance does not establish a conforming Jnana instance.
- **Adjacent work:** *Integrity Enforcement Layer*
  Supplies: Refusal, verification, corruption handling, and effect-bound enforcement.
  Jnana boundary: IEL is an enforcement profile within or beside a realization, not the whole Jnana identity.
- **Adjacent work:** *Universal Governance Kernel*
  Supplies: Compact executable governance-kernel candidate and receipt/certification surfaces.
  Jnana boundary: UGK may realize selected contracts; it does not inherit Jnana-wide conformance by name.
- **Adjacent work:** *Jnana II* and related successors
  Supplies: Alternative or later transformation-substrate formulations.
  Jnana boundary: Their exact version-family and successor relations remain open editorial questions, not automatic merges.

The minimum vertical-slice claim should bind six independently inspectable contracts:

$$V = (Meaning, Law, Lowering, Effect, Receipt, Projection).$$

A slice is internally coherent only when the output identity and obligations of

each contract become declared inputs to the next, and when refusal or semantic remainder cannot be bypassed through another route. This does not prove completeness, security, usefulness, or system-wide conformance.

The GMB supplies durable meaning, sources, standing, conflicts, and revision. Jnana consumes or projects governed meaning through an executable architecture; it is not the GMB itself. The universal GSM should type `REPRESENTED_AS`, `ADMITTED_UNDER`, `LOWERED_TO`, `AUTHORIZED_FOR_EFFECT`, `OBSERVED_BY`, `RECEIPTED_AS`,
`PROJECTED_AS`, and `REVISED_BY`, with exact versions and claim ceilings on every edge.

Corpus integration also adds a conformance rule: **a partial realization names the contracts it implements and must not use the umbrella name as evidence for omitted layers.** A reference implementation may demonstrate one route through the architecture without becoming the architecture's sole definition.

This integration establishes four editorial commitments: adopt layered constitutional containment architecture as Jnana's official corpus role; adopt the six-contract vertical slice; adjudicate the version and successor relations among Jnana, Jnana II, and nearby system families; and define the minimum evidence required before any implementation may claim bounded Jnana conformance.

Admission, Domain Realization, Continuation Lens, and Retrieved Evidence

The fixed-seed comparison with historical CCF, ACG for Quants, Continuation Primacy, and governed retrieval distinguishes four interfaces to Jnana:

$$\begin{aligned} J = (&admitted_meaning, law, lowering, domain_profile,\\ &effect_contract, receipts, projection, evidence_inputs,.\\ &continuation_classification, remainder, revision) \end{aligned}$$

Historical CCF can supply extraction and promotion lineage whose current admitted result resides in the GMB. ACG for Quants can be one domain profile whose data, factor, portfolio, order, and attribution transitions require explicit policy and effect contracts. Neither interface makes the source work a Jnana component by identity.

Continuation Primacy may classify reachable, admissible, viable, and legitimately terminal trajectories as an analytic lens. Jnana mediates selected transitions; it does not prove that continuation is ontologically primary or that persistence is legitimate.

Governed retrieval can supply purpose-specific evidence candidates and claim bindings to law or decision processes. Retrieved material remains subject to

admission, conflict, privacy, revocation, authority, and remainder. The universal GSM can type
`CONSUMES_ADMITTED_GMB_MEANING`,
`REALIZES_DOMAIN_PROFILE`, `MEDIATES_SELECTED_CONTINUATION`,
`ACCEPTS_GOVERNED_EVIDENCE_INPUT`, `REFUSES_UNADMITTED_RETRIEVAL`,
`EMITS_EFFECT_RECEIPT`,
and
`REVISES_CONTAINMENT_CONTRACT`.

This randomized review establishes four editorial commitments: adopt the expanded interface object; preserve CCF-to-GMB admission lineage; treat quant as one bounded domain profile; and integrate continuation and retrieval as lenses or inputs rather than architecture identity or authority.

Layered Containment Without Architectural Absorption

Fifteen retained downstream uses place Jnana across AFS, quantitative development, historical crystallization, crystallized governance, monetary automata, continuation, governance compilation, GOORRT, GCML, Jnana II, open-source collaboration, domain standards, enforcement, quantitative evidence, and governed retrieval. The result is a clear corpus role: Jnana is the layered constitutional containment architecture through which proposals may be represented, admitted, lowered, mediated, observed, refused, and revised. It is not every language, representation, kernel, operating system, or social program that interfaces with it.

$$\begin{aligned} J^* = (&subject, constitution, input_profile,\\ &semantic_representation, law,\\ &procedural_representation, admission, capabilities, \ .\\ &effect_boundary, courts, receipts, refusal, recovery,\\ &emergency, conformance, version, remainder) \end{aligned}$$

Each layer must declare its native claim, source of authority, input and output identity, preservation obligations, loss, refusal behavior, evidence, and bypass boundary. GCNL, GSOL, GCML, MCIR, PEIR, Domain Effects, ReceiptLog, and external governors remain distinct components or profiles. A pipeline diagram does not establish that any particular implementation exists, consumes the selected IC, completely mediates effects, or conforms to the architecture.

Emergency and break-glass operation require antecedent authorization, time and scope limits, independent notice, retained evidence, recovery, and non-retroactive review. Partial implementations must declare the missing layers and exact conformance class. Biological DNA, compiler, and metabolism language remains explanatory metaphor unless a specific structural correspondence is stated and tested.

The universal GSM can type
`INSTANTIATES_JNANA_PROFILE`, `LOWERS_THROUGH_JNANA_LAYER`, `PRESERVES_CONSTITUTIONAL_OBLIGATION`, `REFUSES_JNANA_TRANSITION`, `INVOKES_BREAK_GLASS_PROFILE`, `MEDIATES_DOMAIN_EFFECT`, `RECEIPTS_JNANA_OUTCOME`, and
`DECLARES_PARTIAL_CONFORMANCE`.

This review establishes four editorial commitments: adopt the layered containment record; preserve distinct component and profile identities; require layer-local authority, preservation, loss, evidence, and bypass declarations; and keep metaphor, architecture, implementation, conformance, and social-program standing separate.

17. Failure Modes and Counterevidence

Jnana should weaken or fail under evidence that:

- the law or binding-visibility problem is undecidable in the selected root language;
- PEIR or a compiler can widen meaning, authority, or effects without changing the admitted roots;
- a mutable runtime can rewrite or equivocate about its ReceiptLog;
- an external effect occurs while canonical state records refusal or non-occurrence;
- break-glass authority becomes a routine bypass or cannot be revoked;
- constitutional roots can be changed by ordinary-law authority;
- law version, delegation, revocation, or environment is absent from a decision;
- hash equality is used as semantic equivalence across incompatible versions;
- source meaning or ambiguity is erased during canonicalization;
- affected agents cannot actually discover, refuse, contest, exit, or fork bindings;
- privacy or selective disclosure is incompatible with the required receipts;
- admission becomes centralized censorship or denial of service;
- the biological metaphor hides rather than clarifies operational semantics;
- the cost and latency of the machinery exceed the value of the governed domain;
- simpler capability, transaction, policy, and audit systems provide equivalent assurance;
- implementation candidates disagree on architecture, roots, or transition semantics;
- courts prove only local fixtures while broader conformance is claimed.

The deepest threat is **constitutional simulation**: every layer and receipt is present in name, but ambient authority, mutable roots, evidence substitution, or unverifiable effects make the claimed constraints non-binding.

18. Conformance and Evidence Program

A Jnana conformance program should require:

1. normative schemas for every layer and boundary;
2. independent MCIR parsers, canonicalizers, and test vectors;
3. explicit MCIR-Law decidability and resource bounds;
4. constitutional-root, amendment, epoch, delegation, and revocation tests;
5. compiler-correspondence and semantic-difference tests;
6. capability non-widening and ambient-authority tests;
7. effect success, refusal, timeout, crash, duplication, and ambiguity fixtures;
8. append-only, equivocation, truncation, privilege-separation, and export tests for receipts;
9. deterministic replay where claimed and nondeterminism disclosure elsewhere;
10. projection freshness, partiality, conflict, and reproducibility tests;
11. APD detectability, refusal, contest, exit, renegotiation, and forkability tests;
12. break-glass expiry, recovery, restitution, and post-event review tests;
13. cross-host, cross-runtime, and cross-substrate replay where claimed;
14. composition tests proving obligations survive module and adapter boundaries;
15. one independently reviewed low-consequence vertical slice;
16. claim-ceiling tests that refuse conclusions broader than the evidence.

Evidence should distinguish architecture, specification, implementation candidate, configured instance, executed test, witnessed effect, and published result. None can stand in for the others.

19. Realization Funnel

Stage	Jnana role
Possibility	Explore semantic roots, laws, constitutional regimes, execution models, effect membranes, receipts, projections, authority structures, privacy models, and harmful or unusable configurations.
Specification	Define exact layer schemas, boundary contracts, transition semantics, identities, canonicalization, amendment, effects, receipts, recovery states, conformance, and nonclaims.

Stage	Jnana role
CM1	Enumerate every configuration satisfying the specification and minimum layer obligations, including centralized, brittle, costly, exclusionary, or surveillance-heavy variants.
CM2	Prefer configurations with bounded authority, independent evidence, privacy, recovery, substitution, accessibility, forkability, low loss, and practical negative power.
Instance Configuration	Select exact roots, profiles, laws, actors, authorities, compilers, runtimes, adapters, stores, projections, environments, trust anchors, recovery rules, and tests.
Implementation Realization	Build and run the compiler, court, runtime, effect membrane, receipt system, projections, installation, and recovery mechanisms for that configuration.
Evidence	Execute canonicalization, admission, lowering, effect, ambiguity, replay, security, APD, composition, cross-substrate, and independent-review tests; preserve failures and uncertainty.
Revision	Create successors when meaning, law, constitution, authority, compiler, runtime, effects, evidence, threats, or findings change; preserve epochs and invalidated claims.

The source supplies an architecture and metaphor. The broader corpus contains many implementation and evidence candidates. This paper does not by itself prove that one verified, complete, current Jnana instance exists.

Lawful Semantic Development With an Explicit Remainder

Public orientation

Jnana treats computation as governed semantic development: a proposed meaning or transformation is represented canonically, judged for admissibility, conditionally realized, bounded at effects, and preserved through evidence and revision.

Refusal can protect integrity. It cannot become an excuse for opaque control, permanent stagnation, or architectural claims beyond the demonstrated layer.

Adversarial tests

1. **Canonical-meaning oracle.** Canonical encoding is treated as proof of correct meaning. It establishes reproducible representation under a profile, not truth or adequacy.
2. **Admission-authority capture.** One gate controls what meaning may develop. Law, issuer, jurisdiction, appeal, and plural branches must be explicit.
3. **Refusal-abuse attack.** The system calls every difficult or dissenting proposal inadmissible. Refusal requires reason, scope, evidence, contest, and revision route.
4. **Effect-escape attack.** Semantic admission appears governed while implementation reaches unmediated capabilities. Cross-layer correspondence and effect-boundary tests are required.
5. **Layer-absorption attack.** Jnana is claimed to replace the GMB, compiler, OS, courts, or domain systems. Interfaces and non-identity boundaries must remain explicit.
6. **Conformance-inflation attack.** A partial realization claims the whole architecture. Profiles must state implemented layers, omitted obligations, and evidence.
7. **Liveness failure.** Repeated adjudication prevents lawful work. Budgets, deferral, escalation, and terminal outcomes are constitutional requirements.
8. **Evolution-metaphor overreach.** Semantic variation and selection are presented as biological equivalence or progress. The analogy must not confer empirical or normative standing.

Claim and evidence ceiling

The paper defines a layered constitutional containment architecture and conformance program. It does not establish a universal computation model, complete semantic correctness, implementation maturity, or superiority to other substrates. Source-reported realizations require exact artifact, environment, test, and independent reproduction bindings.

Claims about programming languages, formal methods, security, evolutionary systems, cognition, or performance require external evidence and comparison. Internal corpus relations establish intended composition only. No external research was added in this pass.

Reader-facing consequence

The public promise is computation that can explain which meaning was proposed, under what law it was admitted, how it became an effect, why it was refused or revised, and which parts remain outside the demonstrated boundary.

20. Research Agenda and Open Questions

1. Adjudicate Jnana's exact identity relative to the Governed Semantic Kernel, Jnana II, MosaicOS, and staged implementation families.
2. Approve or revise the seven-concern model: five principal layers plus constitutional invariants and state projection.
3. Define the exact relations among PEIR, GCML, GOML, GSOL, and other machine or operation representations.
4. Define the constitutional root, predecessor-law amendment, successor epoch, fork, and immutability semantics.
5. Define effect ambiguity, commit, reconciliation, compensation, and rollback semantics.
6. Define the enforceable system boundary for APD and negative-power claims.
7. Decide the minimum complete vertical slice that may claim Jnana conformance.
8. Reconcile and evidence-bind the existing implementation candidate families without merging them by name or source-tree proximity.
9. Authorize external research before claims comparing Jnana with biological systems, constitutional theory, capability security, policy engines, effect systems, audit logs, or related computing architectures.

21. Conclusion

Jnana's central insight survives scrutiny: semantic representation, law, execution, external effect, memory, and projection should not collapse into one opaque runtime. Their separation makes it possible to ask not only whether a computation ran, but which meaning was admitted, under what law, with what authority, through which lowering, across which effect boundary, and with what evidence.

The architecture becomes stronger when its metaphors are bounded. MCIR is not omniscient DNA; it is a versioned canonical representation. MCIR-Law is not self-legitimating sovereignty; it is a bounded admission regime. PEIR does not inherit ambient permission. A receipt is not an effect, and an effect is not always reversible. Negative power is not guaranteed by syntax. Partial implementations are valuable when they do not claim the whole.

Jnana is therefore best understood as a candidate architecture for **layered constitutional containment**: expressive computation remains possible, but meaning, authority, effects, and persistence must cross explicit, evidenced boundaries. Refusal may preserve integrity. Deferment and uncertainty may preserve honesty. Canonical state may advance only as far as its evidence allows.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Possibility Space.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.

Corpus-Wide Reference Positioning

This paper argues that jnana is a candidate governed semantic compilation architecture that separates canonical meaning, admissibility law, constitutional amendment constraints, procedural execution, external effects, structural receipts, and derived projections. Its nearest external comparison fields are semantic, governance, security. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Unforgeable capabilities can constrain exercise of authority while still leaving allocation, delegation, revocation, and social legitimacy incorrectly specified. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E2**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

[**E5**] Peter Gardenfors. *Conceptual Spaces: The Geometry of Thought*. MIT Press, 2000. mitpress.mit.edu source **Relationship:** Predecessor geometric treatment of concepts; cited as comparison, not as validation of the paper's manifold terminology.

[**E6**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E7**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

Ableman corpus

[**A1**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-5F976917F0B283EA;
source SHA-256
A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-D95CCC4FB7B58BFB;
source SHA-256
A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity PAPER.PUB.20257244;
source SHA-256
B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-358AC09F5A13587E;
source SHA-256
AFB29CFCB4B58A78B79709AD453CEE9088CCB2AE40F30A70260F1B54F5160579.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Semantic Legitimacy Before Context.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-091E34BA93992CE4;
source SHA-256
DEB7626CE1AB54F4B541347DCC3C4E14DC4F682230ACCCE1A93D1D536A50C44C.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Ableman Constitutional Governance.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-79353AF163A4B0F1;
source SHA-256
FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-D2D97C4042FD281F;
source SHA-256
FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Ableman Corpus references [**MCIR**] Adam Ableman. *MCIR: A Canonical Intermediate Representation for Governed Transformation*. Exact reviewed source SHA-256
762E10A91D94B949F5C862CEB732BE976A0CDF74C828061353AAED4CF5024510;
supplied Zenodo title match, with exact publication identity retained separately.

[**JNANA-II**] Adam Ableman. *Jnana II: Constitutional Transformation Substrate*. Exact reviewed source SHA-256
7F66DCD868229A4DFED7E91DD5A4647A1C54159B77128F464BFF5A9962634AF0;
unpublished source requiring relationship adjudication.

[**TAPD**] Adam Ableman. *Turing Completeness and the Ableman Prime Directive*. Exact reviewed source SHA-256
80F4FCB00B214D22BF0E0A2D67E726DBA4410BAA751300A24DAAD2610E6C395E;
separately custodied reviewed source.

[**GSOL**] Adam Ableman. *GSOL v0.1: Governed Semantic Operation Language*. Exact reviewed source SHA-256
44F8689BF2FD50670CD25347FB2363491115CA7C049C2BBF71D7286A226DD1AC;
supplied Zenodo title match, with exact publication identity retained separately.

[**GCML**] Adam Ableman. *Governed Controlled Machine Language*. Exact reviewed source SHA-256
D4E5D935108BDA55157CADDEAD84B94F14868CEDD5CBF72F5184994983AFFB9A;
unpublished source with unresolved stack placement.

[**RWR**] Adam Ableman. *Reasoning by Way of Receipts*. Exact reviewed source SHA-256
8D884B96597EA6303A324173A7CE2D8C735A952E91FBE4C9EB6509FA6D6B195F;
separately custodied reviewed source.

[**THR**] Adam Ableman. 2026. *The Three Hash Rule*. Zenodo. DOI:
10.5281/zenodo.20788617. Exact reviewed Markdown source SHA-256
03F60317CC9AFB36DDBCF9BF1C6D185611666CC4D9AB290BD4BCC17DF4579B7C.

[**CC**] Adam Ableman. *Constitutional Computing*. Exact reviewed source SHA-256
A1EF0D1F7B5DF84A107E986421EAA95827AEDCB801B7DA63F72BEECA6047BE8D;
supplied Zenodo title match, with exact publication identity retained separately.

[**SS**] Adam Ableman. *Sovereign Software*. Exact reviewed source SHA-256

D8A2707865C6636485A564546D91F2CCF45AAC80161B94AC56A3DA5CF2EB2C13; separately custodied reviewed source.

[CBC] Adam Ableman. *Constraint-Bearing Composability.* Exact reviewed source SHA-256 BEEA0B56DBF7DC5274307E95E08AF0576A9F91A729C1962765361ADA34488B62; supplied Zenodo title match, with exact publication identity retained separately.

No external research was conducted in this pass. Comparisons with biology, constitutional law, compiler architecture, capability security, effect systems, policy engines, audit logs, and related computing architectures require current primary-source verification for a future evidence-qualified revision readiness or novelty claims.

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Zenodo, 2026. doi:10.5281/zenodo.21095303

Adam Ableman *The Three Hash Rule.* Zenodo, 2026. doi:10.5281/zenodo.20788617

Adam Ableman *Governed Semantic Operation Language.* Zenodo, 2026. doi:10.5281/zenodo.21214886

Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus

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DOI: 10.5281/zenodo.23847977